

Practical no. 12

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include<stdio.h>

#include<conio.h>

#include<stdlib.h>

void create();

void insert_at_beg();

void search();

void display();

struct Node{

int data;

struct Node*next;

};

struct Node*head=NULL;

void create(){

int i, n;

struct Node*newNode, *temp;

printf("\nHow many nodes you want to create?:- ");

scanf("%d",&n);

for(i=1;i<=n;i++){

newNode=(struct Node*)malloc(sizeof(struct Node));

printf("\nEnter the data for new node:- ");

scanf("%d",&newNode->data);

newNode->next=NULL;

if(head==NULL){
```

```

head=newNode;

temp=head;

}

else{

temp->next=newNode;

temp=newNode;

}

}

printf("\n%d Nodes are created successfully.",n);

}

void insert_at_beg(){

struct Node *newNode;

newNode=(struct Node*)malloc(sizeof(struct Node));

printf("\nEnter the data for new node:- ");

scanf("%d",&newNode->data);

newNode->next=head;

head=newNode;

printf("\nNode with data %d is successfully inserted at the beginning.",
newNode->data);

}

void search(){

int val,pos=0,found=0;

struct Node*temp=head;

if(head==NULL){

printf("\nLinked list is empty.");

}

printf("\nEnter the data to be searched :- ");

```

```

scanf("%d",&val);

while(temp!=NULL){

if(temp->data==val){

printf("\nNode with data %d is found at position %d",val,pos);

found=1;

break;

}

temp=temp->next;

pos++;

}

if(found==0){

printf("\nNode with %d is not present in the list.",val);

}

}

void display(){

struct Node*temp=head;

if(head==NULL){

printf("\nLinked list is empty.");

}

else{

printf("\nElements of linked list are :- ");

while(temp!=NULL){

printf("\n%d",temp->data);

temp=temp->next;

}

printf("\n<--->");

}

```

```
}

void main(){

int choice;

clrscr();

while(1){

printf("\n*****Linked List*****");

printf("\n1.Create\n2.Insert at beginning\n3.Search\n4.Display\n5.Exit");

printf("\nEnter your choice :- ");

scanf("%d",&choice);

switch(choice){

case 1: create();

        break;

case 2: insert_at_beg();

        break;

case 3: search();

        break;

case 4: display();

        break;

case 5: getch();

        exit(0);

default: printf("\nInvalid Choice!!");

        break;

}

}

getch();

}
```

OUTPUT :-

```
NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles,

*****Linked List*****
1.Create
2.Insert at beginning
3.Search
4.Display
5.Exit
Enter your choice :- 1

How many nodes you want to create?:- 2

Enter the data for new node:- 20

Enter the data for new node:- 30

2 Nodes are created successfully.
```

```
*****Linked List*****
1.Create
2.Insert at beginning
3.Search
4.Display
5.Exit
Enter your choice :- 2

Enter the data for new node:- 10

Node with data 10 is successfully inserted at the beginning.
```

```
*****Linked List*****
1.Create
2.Insert at beginning
3.Search
4.Display
5.Exit
Enter your choice :- 3

Enter the data to be searched :- 30

Node with data 30 is found at position 2
```

```
*****Linked List*****
1.Create
2.Insert at beginning
3.Search
4.Display
5.Exit
Enter your choice :- 4

Elements of linked list are :-
10
20
30
<--->
```